Art Direction

Target look

`docs/reference/level_style_reference.png` (owner's reference, "Twin Towers") is the target: glossy cartoon characters and props with a **thin dark-brown outline (#3a2212)**, painted realm backdrops, warm wood towers on grey stone bases, terracotta tiled roofs with red flags, red banners, a grassy boulder cliff as ground, leafy branches/fence/bushes framing the screen, wooden-sign and dark-slate HUD panels, round orange buttons with labels.

Legal line: we take the *style*, never the *content*. No birds, no green pigs, no Y-shaped slingshot, no Rovio fonts or logos.

Two art pipelines

Asset type	Pipeline
Characters, realm backgrounds	AI images from the owner → `tools/cutout.py` → `assets/art/ai/cut/`
Blocks + damage stages, roofs, flags, banners, catapult, ground cliff, foreground framing, UI, map nodes	SVG from `tools/gen_art.py`

What `tools/cutout.py` does to AI images

1. Moves new files from `assets/art/ai/` to `art_source/ai/` (originals kept, never shipped).
2. Removes the white background by flood fill from the borders, plus large enclosed white pockets (e.g. inside a bow); small whites such as eyes and teeth stay.
3. Soft anti-aliased edge and colour decontamination so no white halo remains; specks removed.
4. Light colour grade, trim, one pixel scale per character family so all poses match in size.
5. Adds the 5 px dark-brown outline used by the rest of the art, then light sharpening.
6. Mirrors archers to face left. Backgrounds are graded and resized to 900 px tall.
7. Writes `meta.json` (feet point, body circle, sizes) used by `Art` to place sprites.

Lighting and shape rules

- Light from top-left. Highlight = lighter tint + white ellipse at 25–40% opacity.
- Characters: ~70% body is one round shape. Eyes take ~30% of face width. Feet and hands are small.
- Blocks: corners rounded 6–10 px at 1x. Wood shows grain, stone shows 2–4 bricks with mortar.
- Silhouettes must read at 64 px tall on a phone.

Palette

Use	Colours
Sky	`#2f9df4` top → `#8fd6ff` → `#e4f7ff` horizon
Grass	`#7ccf3a` light, `#4f9e26` mid, `#2f6e1c` shadow
Wood	`#e0a35c` light, `#b8733a` mid, `#7e4a22` dark
Stone	`#d7dde4` light, `#a7b1bd` mid, `#6f7b89` dark
Iron	`#8a96a3`, `#56616e`, `#2f3842`

Use	Colours
Our team (Vanguard)	blue `#3d7fe0`, gold `#ffc93a`, cream `#fff3d6`
Enemy (Redcaps)	red `#e04040`, maroon `#8e1f2a`, leather `#7a4b2a`
Fire	`#fff3a0` → `#ffb02e` → `#ff5a1f`
UI primary (play/next)	green `#5ccf3c` / `#2f9a28`; secondary orange `#ffae2b` / `#e06a10`; panels cream `#fff4dc` with wood frame

Characters

Character	Description
Rocky (stone)	Round grey boulder with moss tuft on top, heavy brow, grumpy determined eyes, small chip on cheek
Blaze (fire pot)	Round orange clay pot, flame "hair" on top, fierce excited eyes, cork-band belt
Boomer (bomb)	Round navy-black bomb, short lit fuse curling up, huge grin, gold band
Twister (cyclone)	Pale teal swirl-shaped puff, spiral body, dizzy happy eyes, leaves orbiting
Vanguard soldier	Round blue-tunic soldier, gold-trim kettle helmet, small round shield, rosy cheeks, brave smile
Redcap Archer	Round red-hooded archer, pointed hood with feather, mischievous squint, short bow, leather quiver
Iron Baron (boss, later)	Big round armoured man, curled moustache, spiked iron helmet

Current SVG placeholders exist for every character (`tools/gen_art.py`). Each character needs poses: idle, blink (eyes closed), hit/dizzy, and for projectiles: flying (squashed with motion). Soldiers: idle, cheer, duck, knocked out. Archers: idle, draw bow, hit, defeated.

Biomes (8 realms)

1. **Blossom Valley** (gap: water): moonlit or dawn valley, cherry blossom, stone masonry ledges.
2. **Desert Ruins** (water): sandstone mesas, palms, ruined arches, layered red rock.
3. **Pirate Harbor** (sea): sunset or storm harbour, wooden piers on posts.
4. **Crystal Mines** (lava): dark rock with glowing cracks and crystals, mine-cart rails.
5. **Sky Islands** (clouds): floating grassy rocks over a sea of clouds.
6. **Scarlet Highlands** (water): red-orange autumn hills, stone keeps; leaves tinted by `weather_hue`.
7. **Storm Crags** (sea): wet cliffs under storm skies, rain weather, shorter corner trees (`fg_height_frac`).
8. **Iron Citadel** (lava): the Baron's iron fortress city over lava, gears and chains.

Until the owner's AI art for a realm exists, its painting falls back to an older one (`fallback_bg` in `balance.json`) and the stage pieces are SVG placeholders from `gen_art.py` (`stage_assets`).

Each background = 3 parallax layers: far (sky + mountains), mid (hills/landmarks), near (ground decoration, bushes). Backgrounds are delivered sharp and softened in game (downscale + haze shader); the play area (ledge, bases, fortress) stays sharp in front of them, like the reference screenshots.

UI

Rounded panels with a wooden or cream bevel, big rounded buttons with a 6 px darker bottom edge (pressed look), Lilita One for titles with a white-to-yellow gradient and dark brown shadow, Fredoka for body text. Icons are simple filled SVG shapes.

Asset list (phase 1)

`assets/art/svg/`: blocks (wood plank long/short/post, crate, stone square/long, roof, barrel) × 3 damage stages, debris pieces, catapult (base, arm, bucket, wheel), arrow, ground strip, sky gradient + clouds + hills (placeholder background), all 4 heroes, soldier, archer, UI buttons, star, coin, heart, pause/restart icons.